EXPERIMENT 16

Determination and plotting of Radiation Pattern for a uniform linear array (ULA).

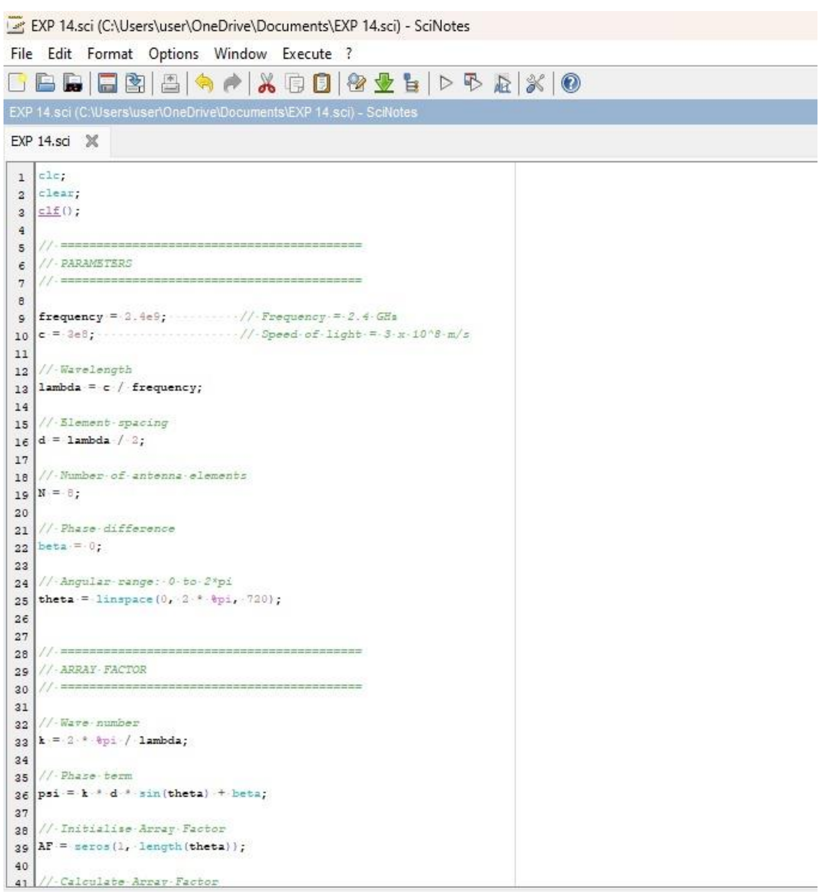
Aim:

To determine and plot the Radiation pattern by Simulating the array factor for a uniform linear array (ULA) using the given parameters and visualize the normalized array factor in polar form.

Software required:

- SCILAB version 6.0.1

Program:



EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

```
clc;
clear;
clf();

// ==========================================
// PARAMETERS
// ==========================================

frequency = 2.4e9;          // Frequency = 2.4 GHz
c = 3e8;                    // Speed of light = 3 x 10^8 m/s

// Wavelength
lambda = c / frequency;

// Element spacing
d = lambda / 2;

// Number of antenna elements
N = 8;

// Phase difference
beta = 0;

// Angular range: 0 to 2*pi
theta = linspace(0, 2 * %pi, 720);


// ==========================================
// ARRAY FACTOR
// ==========================================

// Wave number
k = 2 * %pi / lambda;

// Phase term
psi = k * d * sin(theta) + beta;

// Initialise Array Factor
AF = zeros(1, length(theta));

// Calculate Array Factor
```

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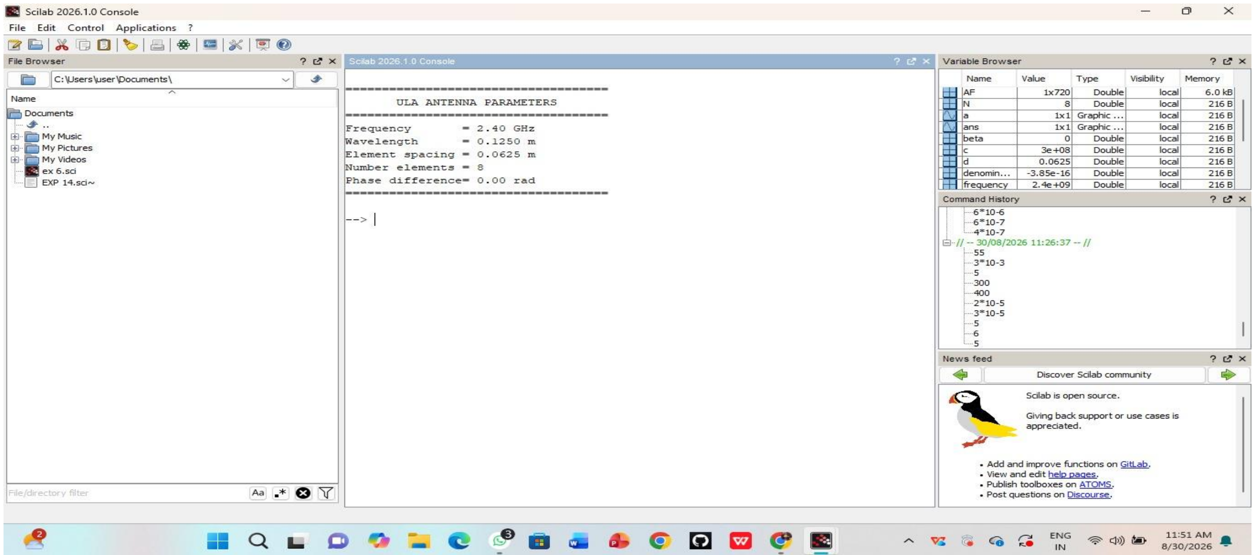
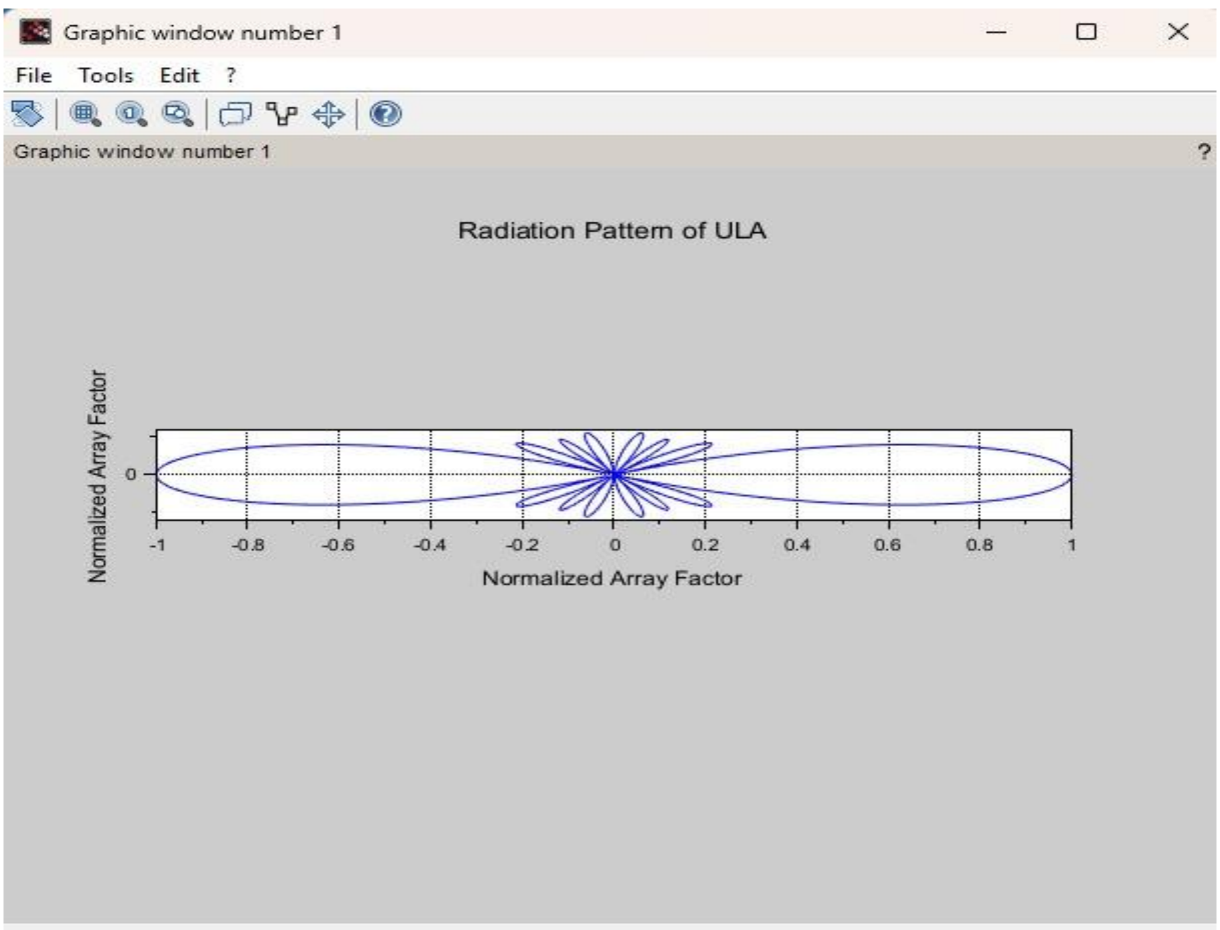
EXP 14.sci

```
// Calculate Array Factor
for i = 1:length(theta)

    denominator = sin(psi(i) / 2);

    // Avoid division by zero
    if abs(denominator) < 1e-10 then
        AF(i) = 1;
    else
        AF(i) = abs(sin(N * psi(i) / 2) / ...
                    (N * denominator));
    end

end


// ==========================================
// NORMALIZATION
// ==========================================

AF = AF / max(AF);


// ==========================================
// CONVERT POLAR TO CARTESIAN
// ==========================================

x = AF .* cos(theta);
y = AF .* sin(theta);


// ==========================================
// RADIATION PATTERN
// ==========================================

figure(1);

plot(x, y);

a = gca();
a.isoview = "on";

xlabel("Normalised Array Factor");
ylabel("Normalised Array Factor");

title("Radiation Pattern of ULA");

xgrid();


// ==========================================
// DISPLAY PARAMETERS
// ==========================================

mprintf("\n");
mprintf("========================================\n");
mprintf("------- ULA ANTENNA PARAMETERS -------\n");
mprintf("========================================\n");
mprintf("Frequency       = %.2f GHz\n", frequency / 1e9);
mprintf("Wavelength      = %.4f m\n", lambda);
mprintf("Element spacing = %.4f m\n", d);
mprintf("Number elements = %d\n", N);
mprintf("Phase difference= %.2f rad\n", beta);
mprintf("========================================\n");
```

Output:


Scilab 2026.1.0 Console
ULA ANTENNA PARAMETERS
Frequency = 2.40 GHz
Wavelength = 0.1250 m
Element spacing = 0.0625 m
Number elements = 8
Phase difference= 0.00 rad


Graphic window number 1
Radiation Pattern of ULA
Normalized Array Factor
Normalized Array Factor

Case Study 1: Optimization of Beamwidth for High-Resolution RADAR Systems

Program:

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EXP 14.sci

```
clc;
clear;
clf();

// =======================================================
// OPTIMIZATION OF BEAMWIDTH FOR HIGH-RESOLUTION RADAR
// =======================================================

// Radar parameters
frequency = 10e9;               // Frequency = 10 GHz
c = 3e8;                        // Speed of light (m/s)
lambda = c / frequency;         // Wavelength (m)

// Antenna array parameters
N = 46;                         // Number of elements
desired_BW = 5;                 // Desired beamwidth in degrees
beta = 0;                       // Phase difference

// Element spacing selected for approximately 5 degree HPBW
d = 0.44 * lambda;

// Angular range
theta = linspace(0, 2 * %pi, 3600);

// Wave number
k = 2 * %pi / lambda;

// =======================================================
// ARRAY FACTOR CALCULATION
// =======================================================

psi = k * d * sin(theta) + beta;

AF = zeros(1, length(theta));

for i = 1:length(theta)

    denominator = sin(psi(i) / 2);

    if abs(denominator) < 1e-10 then
        AF(i) = 1;
```

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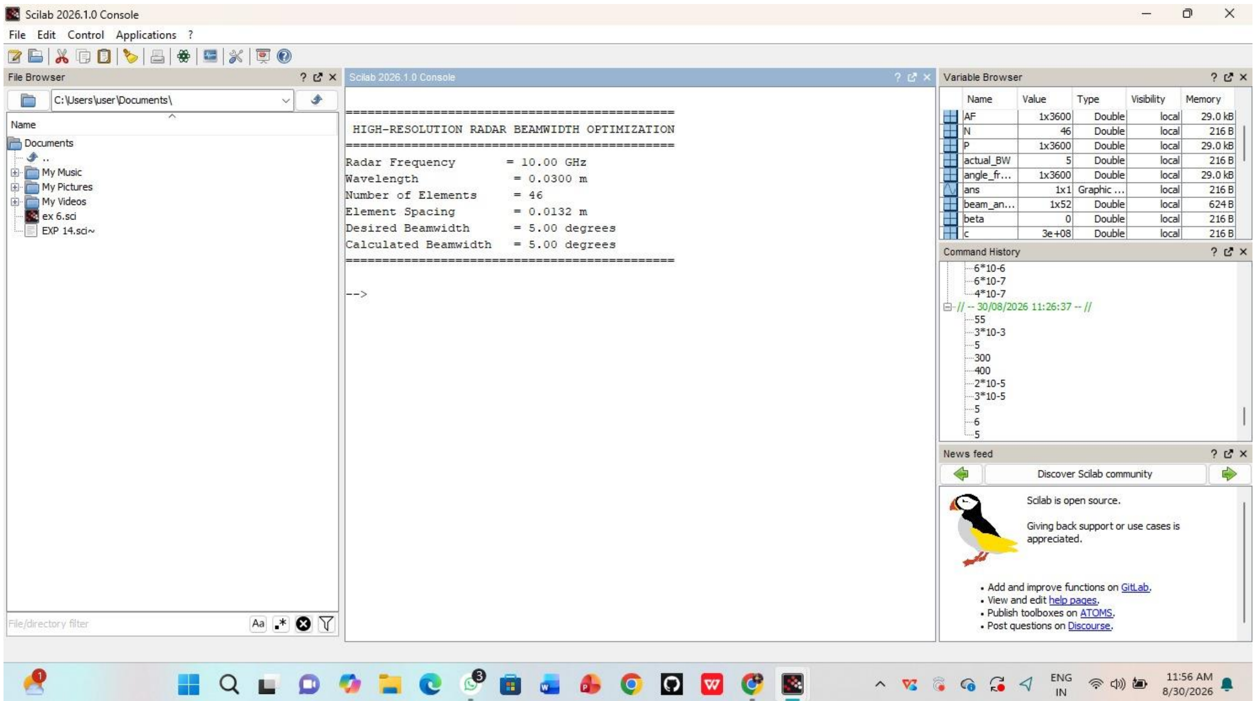
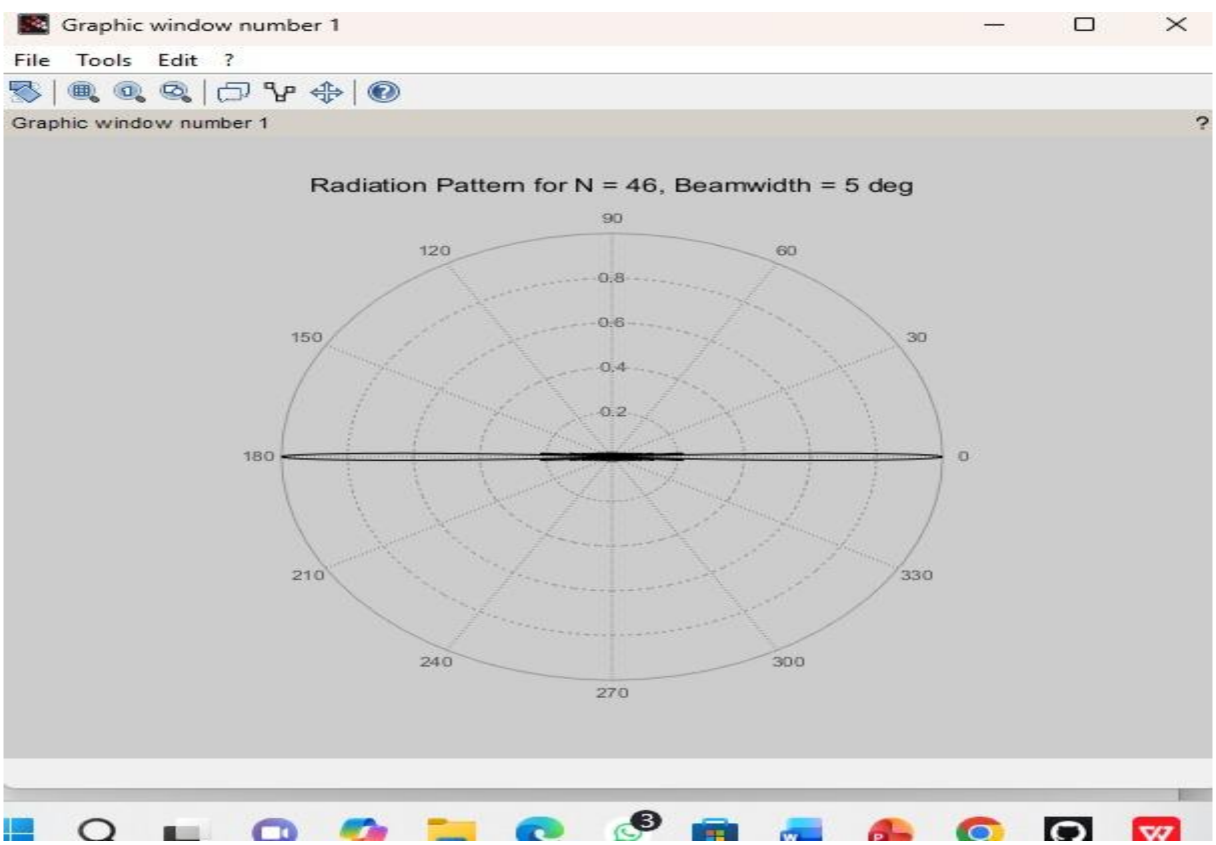
EXP 14.sci

```
    actual_BW = desired_BW;
end


// =======================================================
// CONVERT POLAR COORDINATES
// =======================================================

x = AF .* cos(theta);
y = AF .* sin(theta);


// =======================================================
// POLAR RADIATION PATTERN
// =======================================================

figure(1);

polarplot(theta, AF);

title("Radiation Pattern for N = 46, Beamwidth = 5 deg");


// =======================================================
// DISPLAY RESULTS
// =======================================================

mprintf("\n");
mprintf("==============================================\n");
mprintf(" HIGH-RESOLUTION RADAR BEAMWIDTH OPTIMIZATION\n");
mprintf("==============================================\n");

mprintf("Radar Frequency       = %.2f GHz\n", frequency / 1e9);
mprintf("Wavelength            = %.4f m\n", lambda);
mprintf("Number of Elements    = %d\n", N);
mprintf("Element Spacing       = %.4f m\n", d);
mprintf("Desired Beamwidth     = %.2f degrees\n", desired_BW);
mprintf("Calculated Beamwidth  = %.2f degrees\n", actual_BW);

mprintf("==============================================\n");
```

Output:


Scilab 2026.1.0 Console
==
HIGH-RESOLUTION RADAR BEAMWIDTH OPTIMIZATION
==
Radar Frequency = 10.00 GHz
Wavelength = 0.0300 m
Number of Elements = 46
Element Spacing = 0.0132 m
Desired Beamwidth = 5.00 degrees
Calculated Beamwidth = 5.00 degrees
==
-->


Graphic window number 1
Radiation Pattern for N = 46, Beamwidth = 5 deg
90
120
60
150
30
180
0
210
330
240
300
270
0.8
0.6
0.4
0.2

Case Study 2: End-Fire Radiation for Wireless Communications

Program:

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```
clc;
clear;
clf();

// ========================================================
// END-FIRE AND BROADSIDE RADIATION PATTERN
// FOR WIRELESS COMMUNICATION
// ========================================================

// Basic parameters
frequency = 2.4e9;          // Frequency = 2.4 GHz
c = 3e8;                    // Speed of light (m/s)
lambda = c / frequency;     // Wavelength (m)

// Antenna array parameters
N = 8;                      // Number of elements
d = lambda / 2;             // Element spacing = lambda/2

// Angular range
theta = linspace(0, 2 * %pi, 3600);

// Wave number
k = 2 * %pi / lambda;


// ========================================================
// FUNCTION TO CALCULATE ARRAY FACTOR
// ========================================================

function AF = array_factor(N, k, d, theta, beta)

    psi = k * d * cos(theta) + beta;

    AF = zeros(1, length(theta));

    for i = 1:length(theta)

        denominator = sin(psi(i) / 2);

        if abs(denominator) < 1e-10 then
```

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```
subplot(1,2,2);

plot(x_broadside, y_broadside);
a = gca();
a.isoview = "on";

xgrid();

xlabel("Normalised Pattern");
ylabel("Normalised Pattern");

title("Broadside Configuration (beta = 0)");


// Draw circular reference rings
for r = [0.2 0.4 0.6 0.8 1.0]

    xr = r * cos(theta);
    yr = r * sin(theta);

    plot(xr, yr);

end


// ========================================================
// DISPLAY PARAMETERS
// ========================================================

mprintf("\n");
mprintf("==============================================\n");
mprintf("    ANTENNA ARRAY RADIATION PATTERN\n");
mprintf("==============================================\n");
mprintf("Frequency       = %.2f GHz\n", frequency / 1e9);
mprintf("Wavelength      = %.4f m\n", lambda);
mprintf("Number of Elements = %d\n", N);
mprintf("Element Spacing = %.4f m\n", d);
mprintf("End-Fire Beta   = pi/2 rad\n");
```

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EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

EXP 14.sci

```
// =======================================================
// PLOT BOTH PATTERNS
// =======================================================

figure(1);

// -------------------- END-FIRE --------------------

subplot(1,2,1);

plot(x_endfire, y_endfire);
a = gca();
a.isoview = "on";

xgrid();

xlabel("Normalized Pattern");
ylabel("Normalized Pattern");

title("End-Fire Configuration (beta = pi/2)");


// Draw circular reference rings
for r = [0.2 0.4 0.6 0.8 1.0]

    xr = r * cos(theta);
    yr = r * sin(theta);

    plot(xr, yr);

end


// -------------------- BROADSIDE --------------------

subplot(1,2,2);

plot(x_broadside, y_broadside);
a = gca();
```

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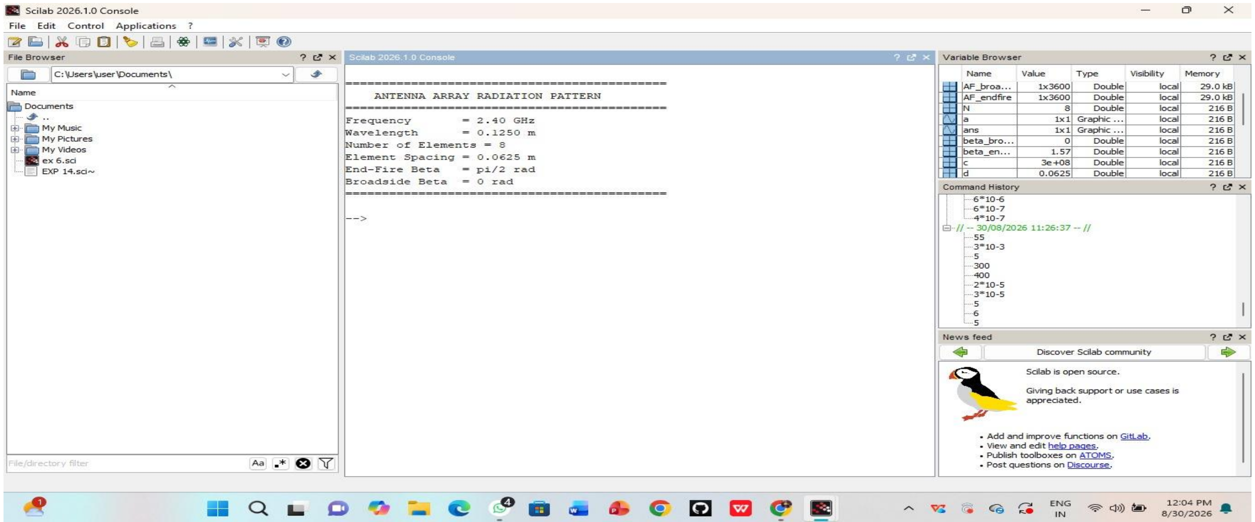
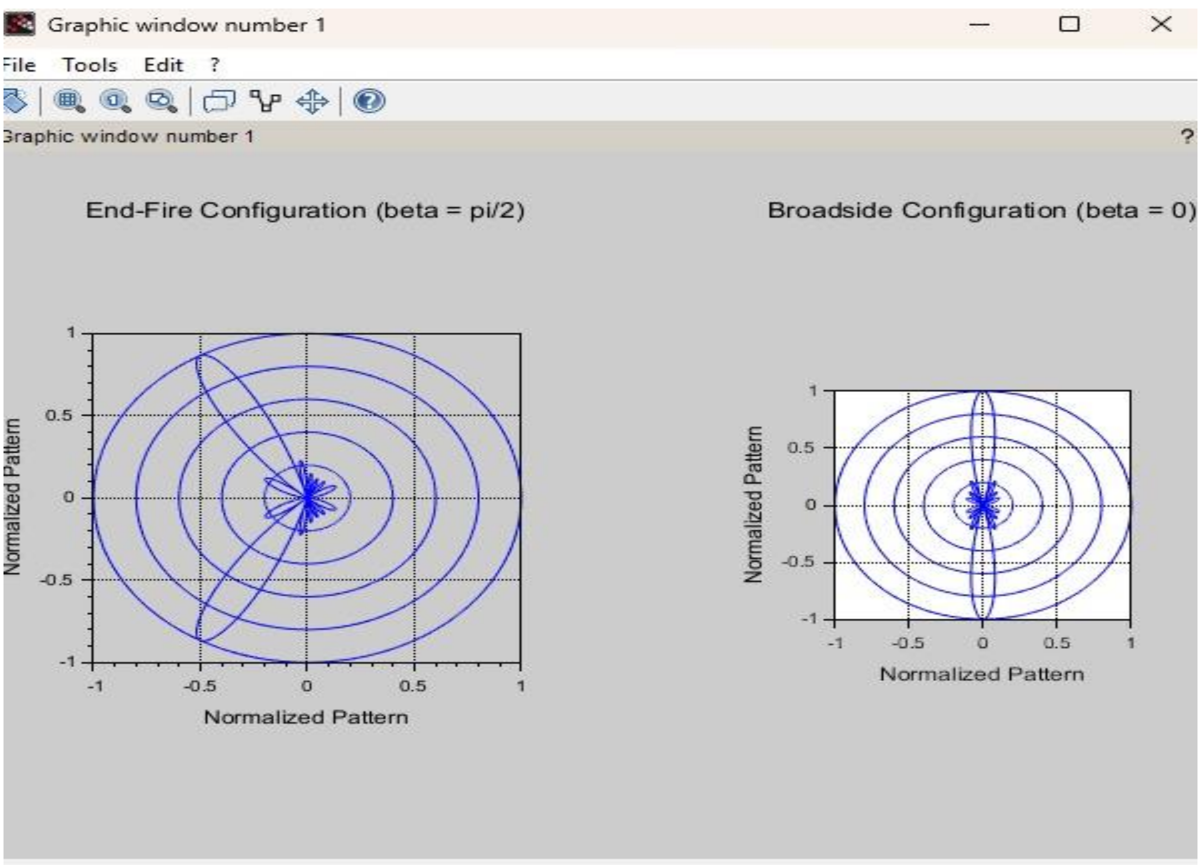
EXP 14.sci

```
    AF = AF / max(AF);

endfunction


// =======================================================
// 1. END-FIRE CONFIGURATION
// beta = pi/2
// =======================================================

beta_endfire = %pi / 2;

AF_endfire = array_factor(N, k, d, theta, beta_endfire);


// =======================================================
// 2. BROADSIDE CONFIGURATION
// beta = 0
// =======================================================

beta_broadside = 0;

AF_broadside = array_factor(N, k, d, theta, beta_broadside);


// =======================================================
// POLAR TO CARTESIAN CONVERSION
// =======================================================

x_endfire = AF_endfire .* cos(theta);
y_endfire = AF_endfire .* sin(theta);

x_broadside = AF_broadside .* cos(theta);
y_broadside = AF_broadside .* sin(theta);


// =======================================================
// PLOT BOTH PATTERNS
// =======================================================
```

Output:


Scilab 2026.1.0 Console
ANTENNA ARRAY RADIATION PATTERN
Frequency = 2.40 GHz
Wavelength = 0.1250 m
Number of Elements = 8
Element Spacing = 0.0625 m
End-Fire Beta = pi/2 rad
Broadside Beta = 0 rad
-->


Graphic window number 1
End-Fire Configuration (beta = pi/2)
Broadside Configuration (beta = 0)
Normalized Pattern
Normalized Pattern
Normalized Pattern
Normalized Pattern

RESULT: The Experiment was done and the radiation characteristics of the ULA is plotted. The different case studies were studied and analysed.